Student Performance Analytics API Documentation

Base URL

Environment	URL
Production	`https://heortappel.pythonanywhere.com`
Local	`http://127.0.0.1:8000`

Authentication

This API uses **Token-based authentication**. Most endpoints require a valid token in the request header.

Header Format

```
Authorization: Token <your_token>
```

Unauthenticated Response

```
{
  "detail": "Authentication credentials were not provided."
}
```

Endpoints

1. User Registration

POST `/api/register/`

Creates a new user account and returns an authentication token.

Authentication: Not required

Request Body:

```
{
  "username": "newuser",
  "email": "user@example.com",
```

```
  "password": "password123"
}
```

Response (201 Created):

```
{
  "message": "User registered successfully",
  "username": "newuser",
  "token": "9a8b7c6d5e4f3g2h1i0j..."
}
```

Error Response (400 Bad Request):

```
{
  "username": ["A user with that username already exists."],
  "password": ["Ensure this field has at least 6 characters."]
}
```

---

2. Obtain Token

POST `/api/token/`

Obtains an authentication token for an existing user.

Authentication: Not required

Request Body:

```
{
  "username": "existinguser",
  "password": "password123"
}
```

Response (200 OK):

```
{
  "token": "9a8b7c6d5e4f3g2h1i0j..."
}
```

Error Response (400 Bad Request):

```
{
  "non_field_errors": ["Unable to log in with provided credentials."]
}
```

---

3. List Students

GET `/api/students/`

Returns a paginated list of all student records.

Authentication: Required

Query Parameters:

Parameter	Type	Description
page	int	Page number (default: 1)
gender	string	Filter by gender (Male/Female)
school_type	string	Filter by school type (Public/Private)
min_score	int	Minimum exam score
max_score	int	Maximum exam score
search	string	Search in gender, school_type, parental_involvement
ordering	string	Sort field: exam_score, hours_studied, attendance, id (prefix - for descending)

Example Request:

```
GET /api/students/?gender=Male&min_score=70&ordering=-exam_score
```

Response (200 OK):


```
{
  "count": 3245,
  "next": "https://heortappel.pythonanywhere.com/api/students/?page=2",
  "previous": null,
  "results": [
    {
      "id": 1,
      "hours_studied": 8,
      "attendance": 95,
      "parental_involvement": "High",
      "access_to_resources": "High",
      "extracurricular_activities": "Yes",
      "sleep_hours": 7,
      "previous_scores": 85,
      "motivation_level": "High",
      "internet_access": "Yes",
      "tutoring_sessions": 2,
      "family_income": "High",
      "teacher_quality": "High",
      "school_type": "Public",
      "peer_influence": "Positive",
      "physical_activity": 4,
      "learning_disabilities": "No",
      "parental_education_level": "Postgraduate",
      "distance_from_home": "Near",
      "gender": "Male",
      "exam_score": 92,
      "batch": 1,
      "batch_number": 1
```

```
    }
  ]
}
```

---

4. Create Student

POST `/api/students/`

Creates a new student record.

Authentication: Required

Request Body:

```
{
  "hours_studied": 8,
  "attendance": 95,
  "parental_involvement": "High",
  "access_to_resources": "High",
  "extracurricular_activities": "Yes",
  "sleep_hours": 7,
  "previous_scores": 85,
  "motivation_level": "High",
  "internet_access": "Yes",
  "tutoring_sessions": 2,
  "family_income": "High",
  "teacher_quality": "High",
  "school_type": "Public",
  "peer_influence": "Positive",
  "physical_activity": 4,
  "learning_disabilities": "No",
  "parental_education_level": "Postgraduate",
  "distance_from_home": "Near",
  "gender": "Male",
  "exam_score": 92
}
```

Response (201 Created):

```
{
  "id": 6608,
  "hours_studied": 8,
  "attendance": 95,
  ...
}
```

---

5. Retrieve Student

GET `/api/students/{id}/`

Retrieves a single student record by ID.

Authentication: Required

Response (200 OK):

```
{
  "id": 1,
  "hours_studied": 8,
  "attendance": 95,
  "parental_involvement": "High",
  ...
  "exam_score": 92,
  "batch": 1,
  "batch_number": 1
}
```

Error Response (404 Not Found):

```
{
  "detail": "Not found."
}
```

---

6. Update Student (Full)

PUT `/api/students/{id}/`

Updates all fields of a student record.

Authentication: Required

Request Body: (Same as Create Student)

Response (200 OK): Updated student object

---

7. Update Student (Partial)

PATCH `/api/students/{id}/`

Updates specific fields of a student record.

Authentication: Required

Request Body:

```
{
  "exam_score": 95,
  "hours_studied": 10
}
```

Response (200 OK): Updated student object

---

8. Delete Student

DELETE `/api/students/{id}/`

Deletes a student record.

Authentication: Required

Response (204 No Content): Empty response

---

9. Student Statistics

GET `/api/students/stats/`

Returns overall statistics and breakdowns by gender and school type.

Authentication: Required

Response (200 OK):

```
{
  "overall": {
    "total_count": 6607,
    "avg_score": 67.24,
    "max_score": 101,
    "min_score": 50,
    "avg_hours_studied": 19.98,
    "avg_attendance": 79.98
  },
  "by_gender": [
    {"gender": "Male", "count": 3245, "avg_score": 67.12},
    {"gender": "Female", "count": 3362, "avg_score": 67.35}
  ],
  "by_school_type": [
    {"school_type": "Public", "count": 3312, "avg_score": 67.18},
    {"school_type": "Private", "count": 3295, "avg_score": 67.30}
  ]
}
```

---

10. Upload CSV

POST `/api/students/upload_csv/`

Uploads a CSV file to import student data with batch tracking.

Authentication: Required

Content-Type: `multipart/form-data`

Request:

```
file: <CSV file>
```

Response (201 Created):

```
{
  "message": "CSV imported successfully",
  "batch_number": 2,
  "record_count": 6607,
  "filename": "StudentPerformanceFactors.csv"
}
```

CSV Format:

```
Hours_Studied,Attendance,Parental_Involvement,Access_to_Resources,Extracurricular_Activities,
8,95,High,High,Yes,7,85,High,Yes,2,High,High,Public,Positive,4,No,Postgraduate,Near,Male,92
```

---

11. List Batches

GET `/api/batches/`

Returns all import batches.

Authentication: Required

Response (200 OK):

```
[
  {
    "id": 1,
    "batch_number": 1,
    "created_at": "2026-03-12T10:30:00Z",
    "record_count": 6607,
    "filename": "StudentPerformanceFactors.csv"
  }
]
```

---

12. Retrieve Batch

GET `/api/batches/{id}/`

Retrieves details of a specific batch.

Authentication: Required

Response (200 OK):

```
{
  "id": 1,
  "batch_number": 1,
  "created_at": "2026-03-12T10:30:00Z",
  "record_count": 6607,
  "filename": "StudentPerformanceFactors.csv"
}
```

---

13. Batch Students

GET `/api/batches/{id}/students/`

Returns all students belonging to a specific batch.

Authentication: Required

Response (200 OK):

```
[
  {
    "id": 1,
    "hours_studied": 8,
    "attendance": 95,
    ...
  }
]
```

---

14. Batch Statistics

GET `/api/batches/{id}/stats/`

Returns statistics for a specific batch.

Authentication: Required

Response (200 OK):

```
{
  "batch_number": 1,
  "stats": {
```

```
    "total_count": 6607,
    "avg_score": 67.24,
    "max_score": 101,
    "min_score": 50,
    "avg_hours_studied": 19.98,
    "avg_attendance": 79.98
  }
}
```

---

15. Batch Analysis

GET `/api/batches/{id}/analysis/`

Returns score distribution (per 10 points) and factor correlation analysis for a batch.

Authentication: Required

Response (200 OK):

```
{
  "batch_number": 1,
  "total_students": 6607,
  "score_distribution": {
    "0-9": 0,
    "10-19": 0,
    "20-29": 0,
    "30-39": 0,
    "40-49": 0,
    "50-59": 831,
    "60-69": 2694,
    "70-79": 2029,
    "80-89": 879,
    "90-100": 174
  },
  "factor_correlations": {
    "hours_studied": 0.1631,
    "attendance": 0.1578,
    "tutoring_sessions": 0.1012,
    "sleep_hours": 0.0534,
    "physical_activity": 0.0198
  },
  "most_impactful_factor": {
    "name": "hours_studied",
    "correlation": 0.1631,
    "interpretation": "weak positive correlation"
  }
}
```

Factor Correlation Interpretation:

- **Strong** (|r| ≥ 0.7): Strong relationship

- **Moderate** (0.4 ≤ |r| < 0.7): Moderate relationship
- **Weak** (0.2 ≤ |r| < 0.4): Weak relationship
- **Very Weak** (|r| < 0.2): Very weak or no relationship

Error Codes

Status Code	Description
200	OK - Request successful
201	Created - Resource created
204	No Content - Resource deleted
400	Bad Request - Invalid input
401	Unauthorized - Authentication required
404	Not Found - Resource not found
405	Method Not Allowed - Invalid HTTP method

Example Usage with cURL

Register User

```
curl -X POST https://heortappel.pythonanywhere.com/api/register/ \
  -H "Content-Type: application/json" \
  -d '{"username":"testuser","email":"test@example.com","password":"test123456"}'
```

Get Token

```
curl -X POST https://heortappel.pythonanywhere.com/api/token/ \
  -d "username=testuser&password=test123456"
```

List Students

```
curl https://heortappel.pythonanywhere.com/api/students/ \
  -H "Authorization: Token YOUR_TOKEN"
```

Create Student

```
curl -X POST https://heortappel.pythonanywhere.com/api/students/ \
  -H "Authorization: Token YOUR_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"hours_studied":10,"attendance":90,"exam_score":85,"gender":"Male","school_type":"Publ
```

Upload CSV

```
curl -X POST https://heortappel.pythonanywhere.com/api/students/upload_csv/ \
  -H "Authorization: Token YOUR_TOKEN" \
  -F "file=@StudentPerformanceFactors.csv"
```

Get Analysis

```
curl https://heortappel.pythonanywhere.com/api/batches/1/analysis/ \
  -H "Authorization: Token YOUR_TOKEN"
```